

DAY 01 — CYBERSECURITY FUNDAMENTALS

13-Day Offline Cybersecurity Program | Professional Learning Documentation

1. Learning Objectives

- Understand the purpose and scope of cybersecurity and information security.
- Build foundational knowledge of operating systems, servers, storage, and enterprise infrastructure.
- Understand networking fundamentals including the OSI model, IP addressing, subnetting, routing, ports, and protocols.
- Identify major cybersecurity domains and their relevance to enterprise environments.
- Understand Red Team, Blue Team, and Purple Team responsibilities.
- Identify foundational technical knowledge expected from a cybersecurity engineer.

2. Cybersecurity & Information Security

Cybersecurity is the practice of protecting systems, networks, applications, devices, and digital assets from unauthorized access, misuse, disruption, and attacks.

Information Security focuses on protecting information and information-processing resources. The core objectives are commonly represented by the **CIA triad: Confidentiality, Integrity, and Availability**.

Hacking refers broadly to techniques used to access, modify, analyze, or interact with systems. In professional cybersecurity, offensive testing must be authorized and performed within an agreed scope.

3. Operating Systems

Operating systems provide the software layer through which users and applications interact with hardware and system resources. Security professionals need OS knowledge to understand processes, services, permissions, authentication, logs, patching, and system hardening.

Category	Examples / Role	Security Relevance
Client OS / End-User OS	Windows, Linux, macOS, etc.	User-facing operating systems; require patching, access control, security configuration, and monitoring.
Server OS	Windows Server, Linux server distributions, etc.	Hosts business and network services; hardening, least privilege, logging, and monitoring are important.

4. Common Server Types

- **Web Server:** delivers web content and HTTP/HTTPS services.
- **Web Application Server:** runs application logic, APIs, and backend services.
- **Database Server:** stores and processes application data.
- **Email Server:** handles email delivery, storage, and related services.
- **Gaming Server:** hosts multiplayer game services and sessions.
- **Load Balancer:** distributes incoming traffic across backend systems.
- **Proxy Server:** acts as an intermediary between clients and destination services.
- **Reverse Proxy:** sits in front of backend services and can provide routing, TLS termination, caching, or access control.
- **Network Server:** provides infrastructure services such as DNS, DHCP, authentication, or other network functions.
- **File Server:** provides centralized file storage and access.

5. Major Cybersecurity Domains

- **Physical Security** — protects facilities, devices, equipment, and physical access.
- **Network Security** — protects network infrastructure, traffic, and communication paths.
- **Application Security** — addresses vulnerabilities in software, APIs, and application components.
- **Identity & Access Management (IAM)** — manages identities, authentication, authorization, and privileges.
- **Cloud Security** — protects cloud workloads, identities, configurations, data, and services.
- **Data Security** — protects data using access control, encryption, classification, backup, and monitoring.
- **Email Security** — addresses phishing, spoofing, malicious attachments, and email-based threats.
- **AI Security** — addresses security risks involving AI systems, models, data, and AI-enabled applications.
- **OS Security** — hardens operating systems and controls privileges, services, and system activity.
- **Hardware / Firmware Security** — protects physical components and low-level software.
- **Container / Embedded Security** — protects containerized workloads and embedded devices.

6. Data, Storage & Infrastructure Security

The session introduced both physical and digital storage environments and their security considerations.

Area	Examples	Security Focus
Physical storage	NAS, SAN, file servers, pen drives, hard disks	Access control, physical protection, backup, availability, and data protection.
Cloud / digital storage	DigiLocker, OneDrive, Google Drive, virtual file services, Amazon S3	Identity, permissions, encryption, logging, secure configuration, and data governance.

GRC: Governance, Risk, and Compliance provides a structured approach for managing security governance, identifying and treating risks, and meeting applicable requirements.

OT / ICS Security: Operational Technology and Industrial Control Systems security focuses on protecting industrial environments such as PLCs, SCADA, DCS, manufacturing systems, and related operational infrastructure.

7. Ethical Hacking Team Models

Team	Primary Focus	Typical Activities
Red Team	Offensive security	Authorized adversary simulation, penetration testing, attack-path validation.
Blue Team	Defensive security	Monitoring, detection, response, hardening, threat analysis.
Purple Team	Collaboration	Combines offensive and defensive insights to improve detection and validation.

8. Foundational Skills for a Cybersecurity Engineer

- Operating-system fundamentals
- Networking fundamentals
- Scripting / programming fundamentals
- Basic virtualization concepts
- Cybersecurity principles including CIA and related controls
- Database fundamentals
- DNS fundamentals
- Digital certificates and PKI fundamentals
- CDN fundamentals
- Web-server fundamentals
- Load balancing and auto-scaling concepts

9. Networking Fundamentals

Network: a group of interconnected computing or electronic devices that communicate and share resources.

OSI Model: a seven-layer conceptual model used to understand how network communication functions.

Layer	Name	Security / Networking Context
7	Application	Application services and protocols such as HTTP, DNS and SMTP.
6	Presentation	Data representation, encoding, and encryption-related functions.
5	Session	Session establishment and management.
4	Transport	End-to-end communication; TCP/UDP and port numbers.
3	Network	Logical addressing and routing; IP.
2	Data Link	Frames, MAC addressing, and switching.
1	Physical	Signals, cabling, radio, and physical transmission.

10. IP Addressing

An IP address is a logical address used to identify a network interface or host for communication. The training introduced IPv4, which uses 32-bit addresses, and the distinction between public and private addressing.

Public IP: globally routable addressing used for communication across the public Internet. **Private IP:** addressing used inside private networks and commonly translated through NAT when accessing public networks.

Private IPv4 ranges: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

Classful IPv4 addressing: the session introduced historical Classes A–E. Modern networks generally use CIDR rather than classful addressing.

Class	First-Octet Range	Traditional Purpose
A	1–126	Large networks
B	128–191	Medium-sized networks
C	192–223	Smaller networks
D	224–239	Multicast
E	240–255	Experimental / reserved

11. Subnetting & Supernetting

Subnetting divides a larger IP network into smaller logical networks. It supports address organization, segmentation, routing boundaries, and security design.

Supernetting combines multiple contiguous networks into a larger address block and is commonly associated with route aggregation and CIDR.

Subnet mask / CIDR prefix: identifies the network and host portions of an IP address. CIDR notation expresses the prefix length, for example /24.

12. Routing & Routing Protocols

Routing is the process of selecting paths for traffic between networks. Routers use routing information to determine where packets should be forwarded.

Protocol	Category / Note
RIP v1	Distance-vector interior routing protocol; legacy protocol.
RIP v2	Updated RIP version supporting CIDR and additional routing information.
EIGRP	Interior gateway routing protocol associated with Cisco environments.

Protocol	Category / Note
IGRP	Legacy Cisco interior gateway routing protocol.
BGP	Inter-domain routing protocol used to exchange routing information between autonomous systems and across the Internet.

13. Ports & Port Number Ranges

A port number is a logical identifier used with transport-layer protocols such as TCP and UDP to direct traffic to the appropriate application or service on a host.

Range	Category	Typical Use
0–1023	Well-known ports	Standard / common services.
1024–49151	Registered ports	Applications and services registered for specific uses.
49152–65535	Dynamic / private / ephemeral	Often used for temporary client-side connections.

14. Load Balancing & Auto Scaling

Load balancing distributes incoming traffic across multiple servers or service instances to improve availability, performance, and resilience.

Auto scaling adjusts compute capacity according to demand or defined policies. These mechanisms help services maintain availability as workloads change.

15. Additional Internet & Web Concepts

- **DNS:** translates domain names into IP addresses and supports service discovery and Internet communication.
- **Digital Certificates / PKI:** support identity verification and encrypted communication such as TLS.
- **CDN:** distributes content through geographically distributed edge locations to improve performance and availability.
- **Web Server:** receives and processes web requests, commonly using HTTP/HTTPS.

16. Assignment / Practical Learning Task

Understand what happens when a user enters a URL into a browser, including the interaction between DNS, IP addressing, routing, network communication, and the destination web server. This forms an important foundation for understanding web and network security.

17. Day 1 Professional Takeaway

Day 1 established the systems and networking foundation required for cybersecurity. The learning progressed from cybersecurity fundamentals and security domains to operating systems, servers, storage, ethical hacking roles, OSI networking, IPv4 addressing, subnetting, routing, ports, and core Internet infrastructure.

These concepts provide the baseline for understanding how security controls are applied across clients, end-user systems, networks, applications, cloud environments, and enterprise infrastructure.

18. Skills / Knowledge Gained

- Cybersecurity and information security fundamentals
- Client OS / End-User OS and server fundamentals
- Network architecture and OSI model
- IPv4 addressing and public/private IP concepts
- Subnetting and supernetting fundamentals
- Routing and routing-protocol awareness
- TCP/UDP port concepts

- Server and infrastructure security awareness
- Cloud, data, GRC, OT/ICS, and security-domain awareness
- Ethical hacking team concepts
- Foundational cybersecurity engineering knowledge

Documentation note: This PDF is a professionally formatted version of the Day 1 training notes. Terminology has been standardized where appropriate for industry usage while preserving the subjects covered during the session.